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FACULTY OF ENGINEERING
AND ARCHITECTURE

Satellite Precoding Techniques

Intermediate Thesis Presentation

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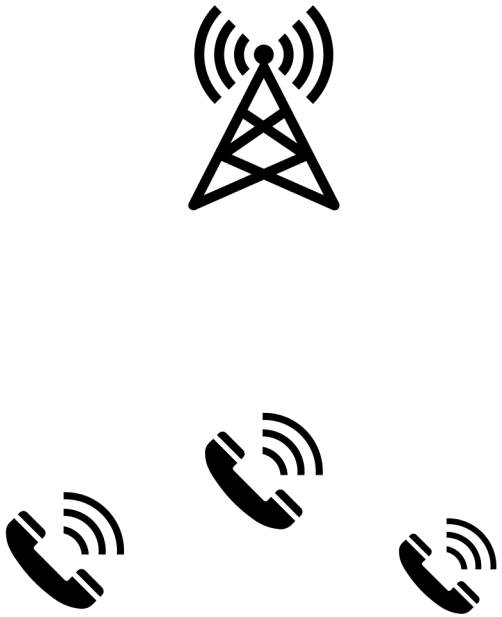
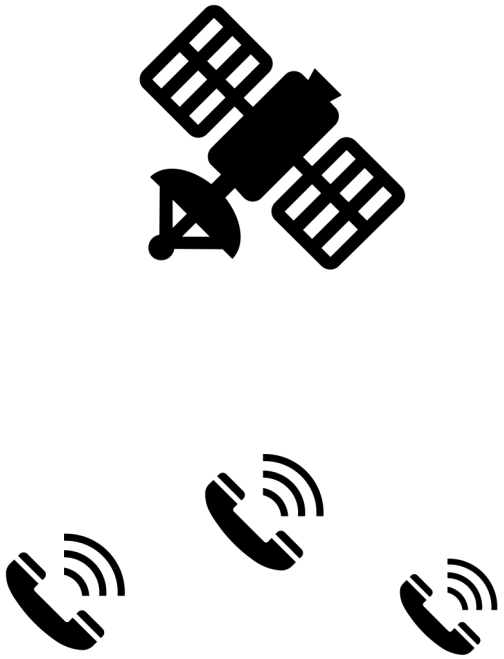
Plan of Approach

1. SU-MIMO

2. MU-MIMO



3. MU-MIMO
with satellite conditions



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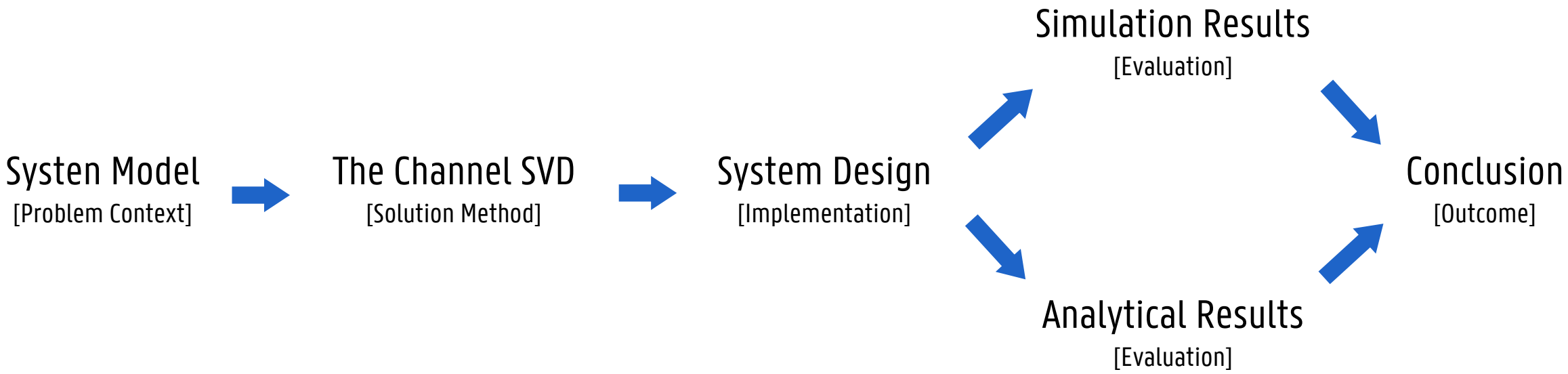
SU-MIMO DigCom System

Overview



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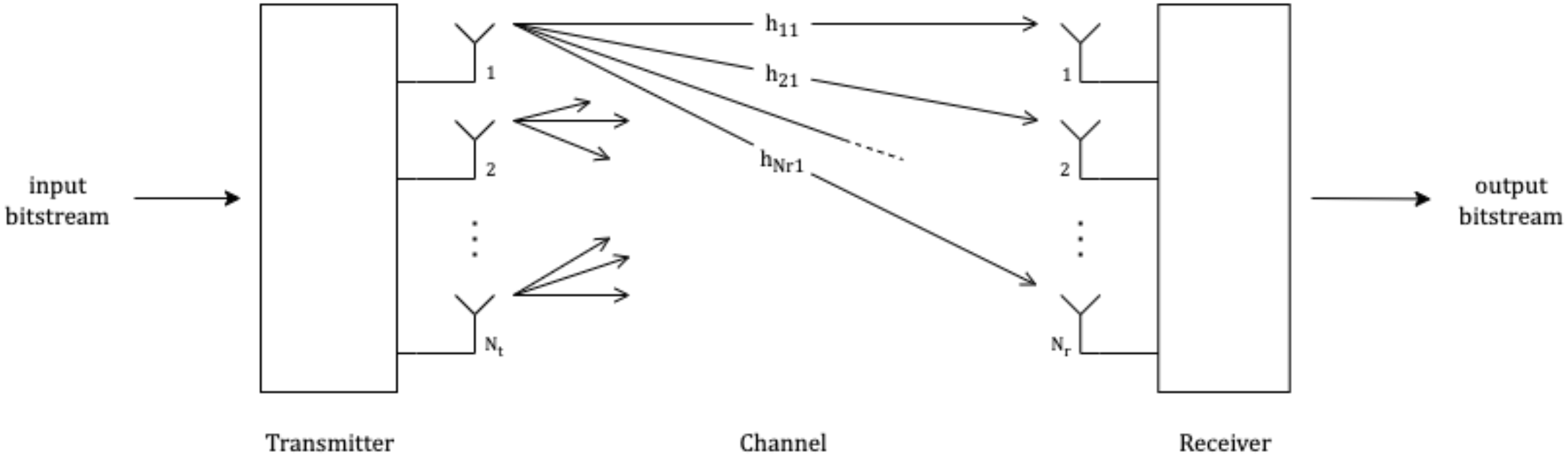
System Model



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The Channel SVD

A Decomposition into R_H parallel eigenchannels.

- The received signal $\mathbf{y}$ can be written in terms of the transmitted signal $\mathbf{x}$ as:

$$\mathbf{y} = \mathbf{H} \cdot \mathbf{x} + \mathbf{w}$$

- The Singular-Value-Decomposition (SVD) of the channel matrix $\mathbf{H}$:

$$\mathbf{H} = \mathbf{U} \cdot \mathbf{S} \cdot \mathbf{V}^{\mathrm{H}}$$

- Precoding at the transmitter and combining at the receiver:

Precoding: $\mathbf{x} = \mathbf{V} \cdot \tilde{\mathbf{x}}$

Combining: $\tilde{\mathbf{y}} = \mathbf{U}^{\mathbf{H}} \cdot \mathbf{y}$

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The Channel SVD

A Decomposition into R_H parallel eigenchannels.

$$\mathbf{y} = \mathbf{H} \cdot \mathbf{x} + \mathbf{w}$$

$$\Leftrightarrow \mathbf{y} = (\mathbf{U} \cdot \mathbf{S} \cdot \mathbf{V}^{\mathrm{H}}) \cdot \mathbf{x} + \mathbf{w}$$

$$\Leftrightarrow \tilde{\mathbf{y}} = \mathbf{U}^{\mathbf{H}} \cdot ((\mathbf{U} \cdot \mathbf{S} \cdot \mathbf{V}^{\mathrm{H}}) \cdot (\mathbf{V} \cdot \tilde{\mathbf{x}}) + \mathbf{w})$$

$$\Leftrightarrow \tilde{\mathbf{y}} = \mathbf{U}^{\mathbf{H}} \cdot \mathbf{U} \cdot \mathbf{S} \cdot \mathbf{V}^{\mathrm{H}} \cdot \mathbf{V} \cdot \tilde{\mathbf{x}} + \mathbf{U}^{\mathbf{H}} \cdot \mathbf{w}$$

$$\Leftrightarrow \tilde{\mathbf{y}} = \mathbf{S} \cdot \tilde{\mathbf{x}} + \mathbf{U}^{\mathbf{H}} \cdot \mathbf{w}$$

$$\boxed{\tilde{\mathbf{y}} = \mathbf{S} \cdot \tilde{\mathbf{x}} + \tilde{\mathbf{w}}}$$

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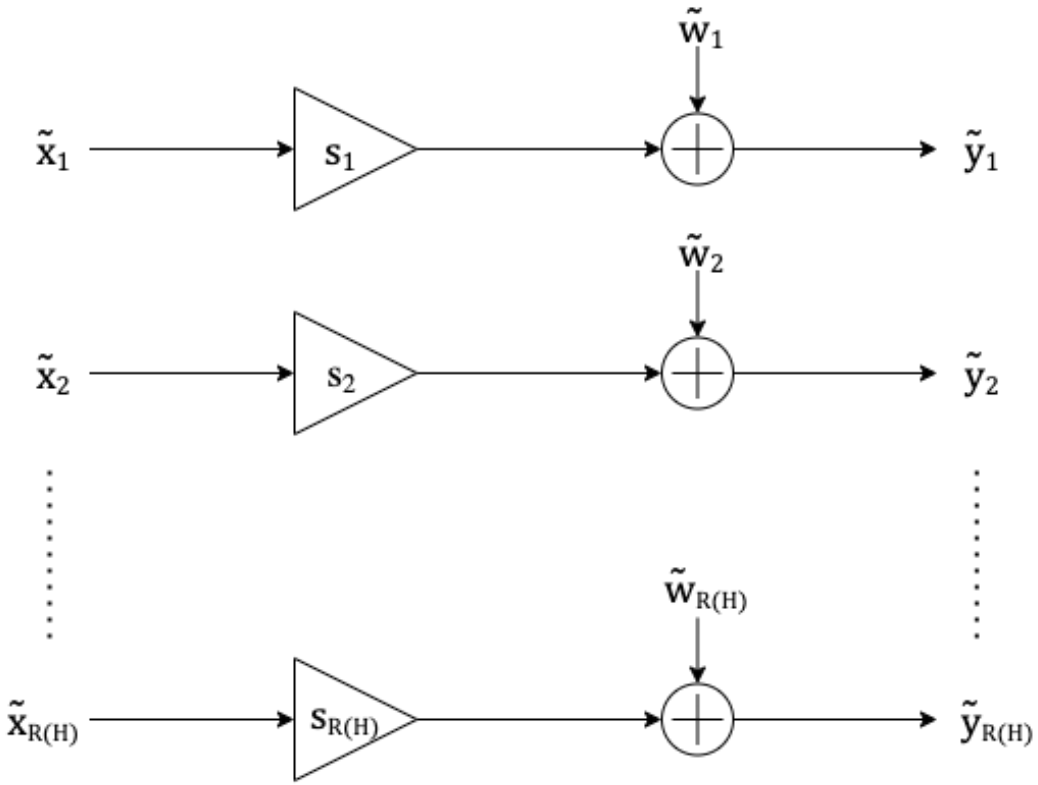
The Channel SVD

A Decomposition into R_H parallel eigenchannels.



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Resource Allocation

How are power and bits allocated across the antennas?

Power Allocation

Equal
- No Water-filling Algorithm
- No CSI required

Eigenbeamforming
- No Water-filling Algorithm
- No CSI Required

Optimal
- Water-filling Algorithm
- CSI Required

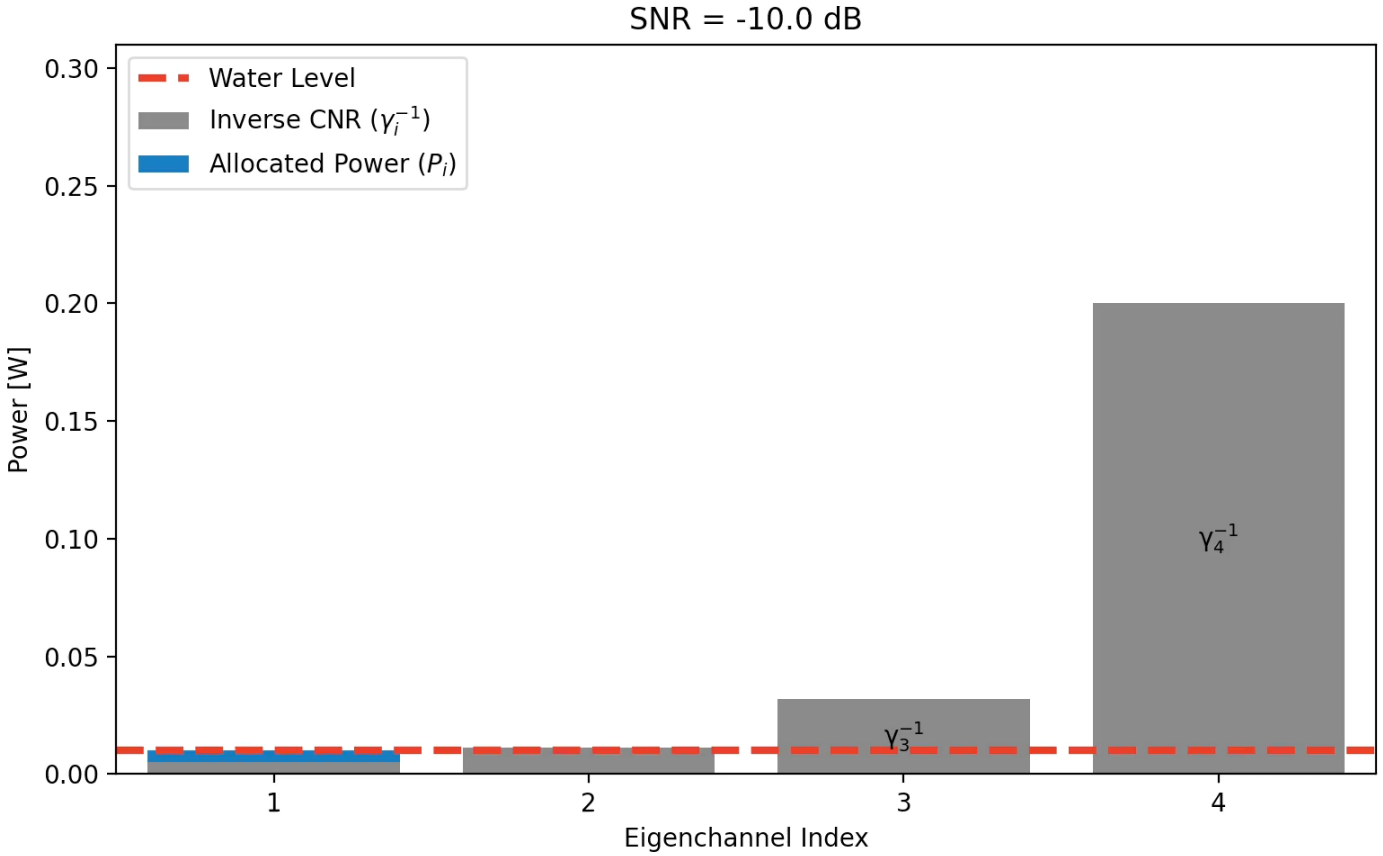
Bit Allocation

Fixed
- Predetermined

Adaptive
- Eigenchannel Capacity

Resource Allocation

Optimal Power Allocation - A Visualisation of the Waterfilling Algorithm



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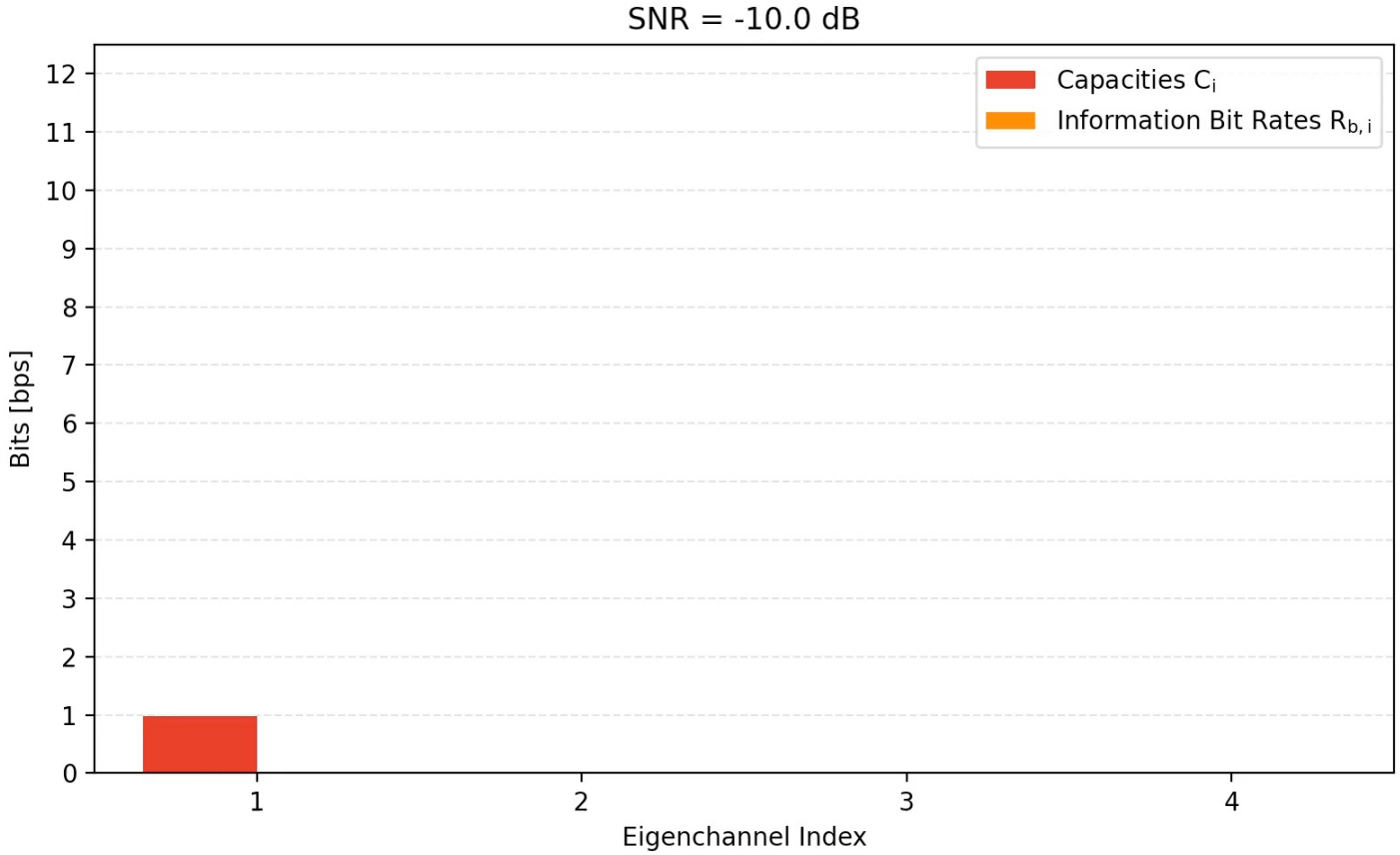
Resource Allocation

Adaptive Bit Allocation - A Visualisation of Information Bit Rates $R_{b,i}$



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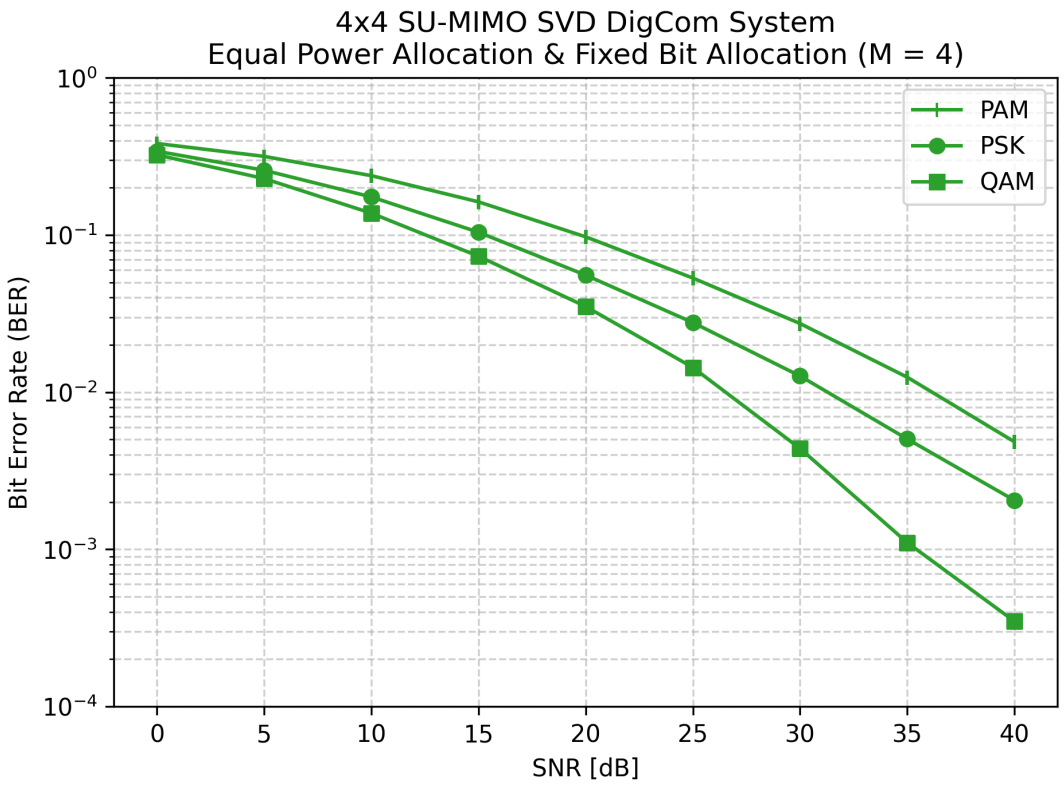
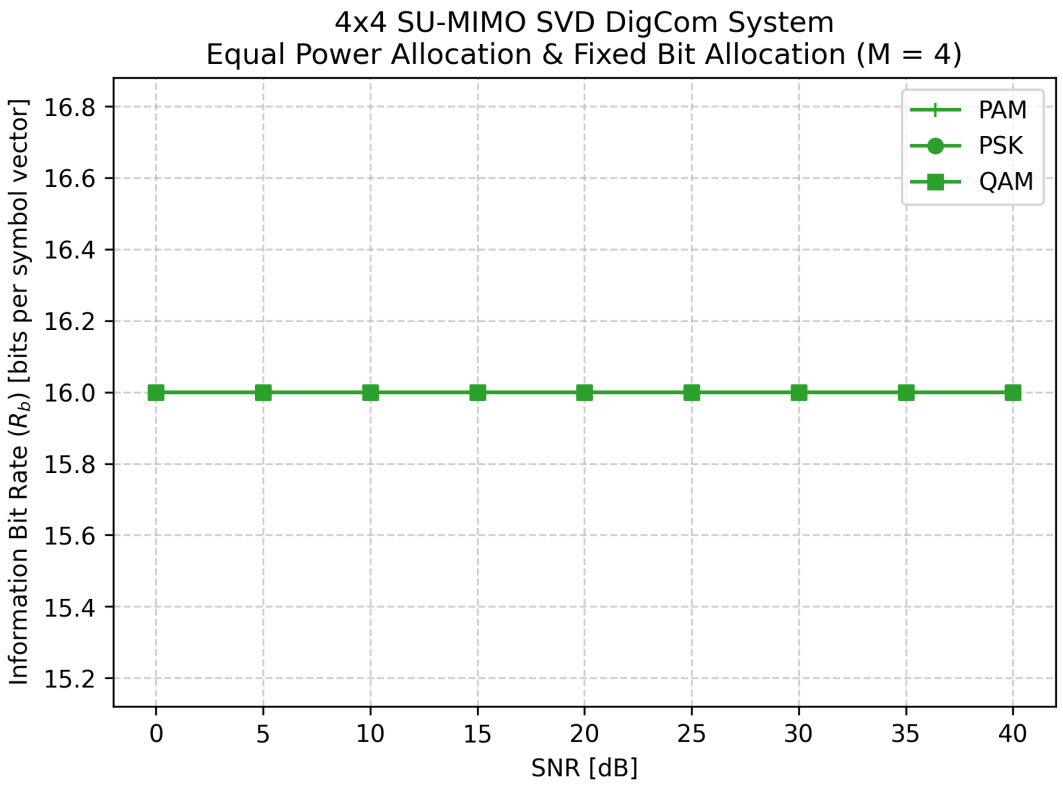
Simulation Results

Equal Power Allocation & Fixed Bit Allocation





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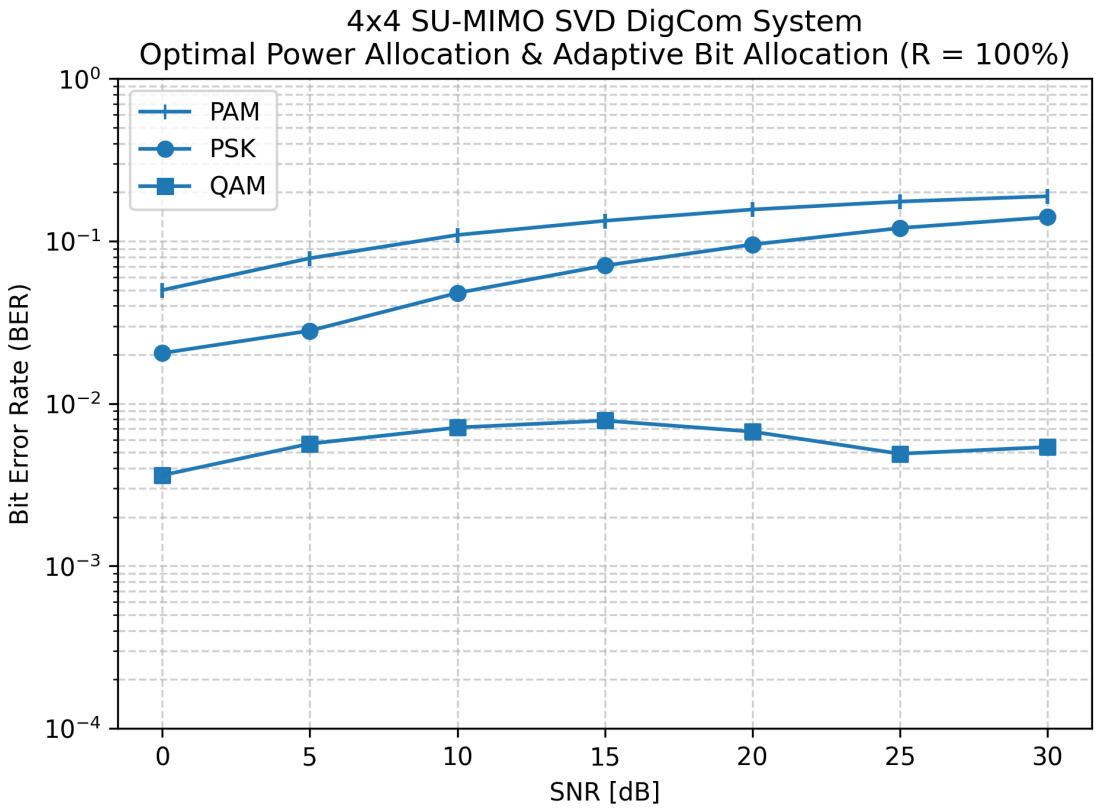
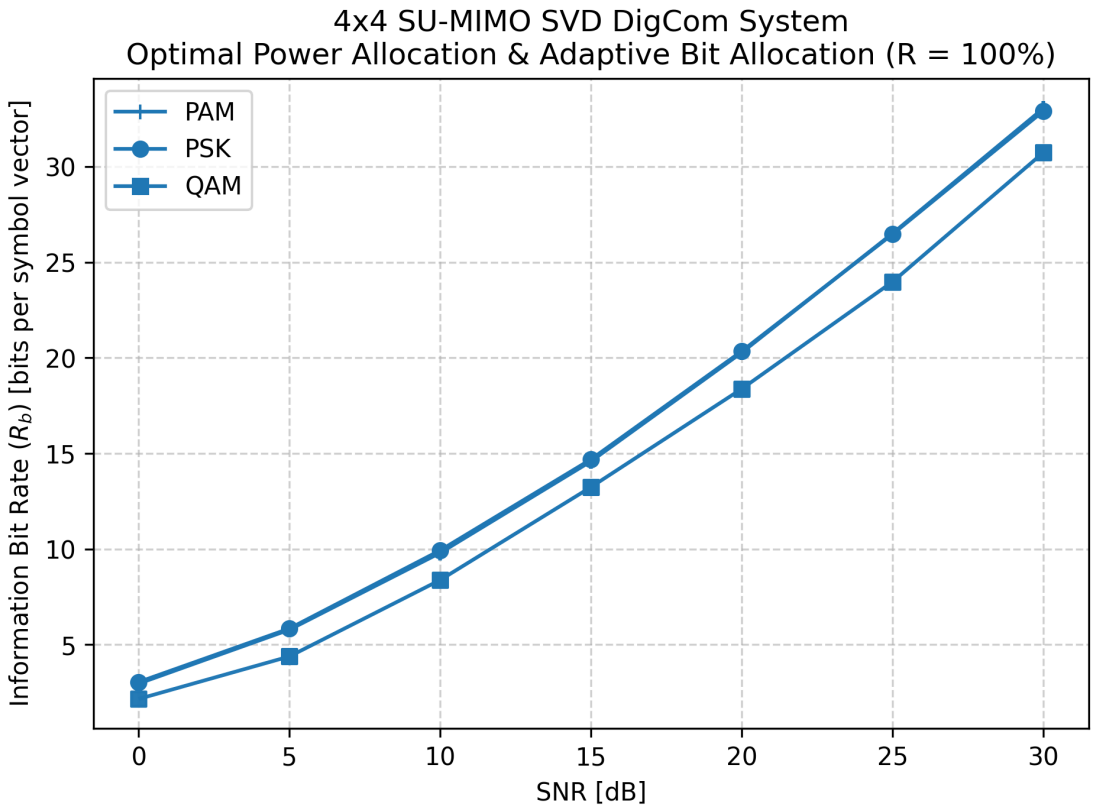
Simulation Results

Optimal Power Allocation & Adaptive Bit Allocation





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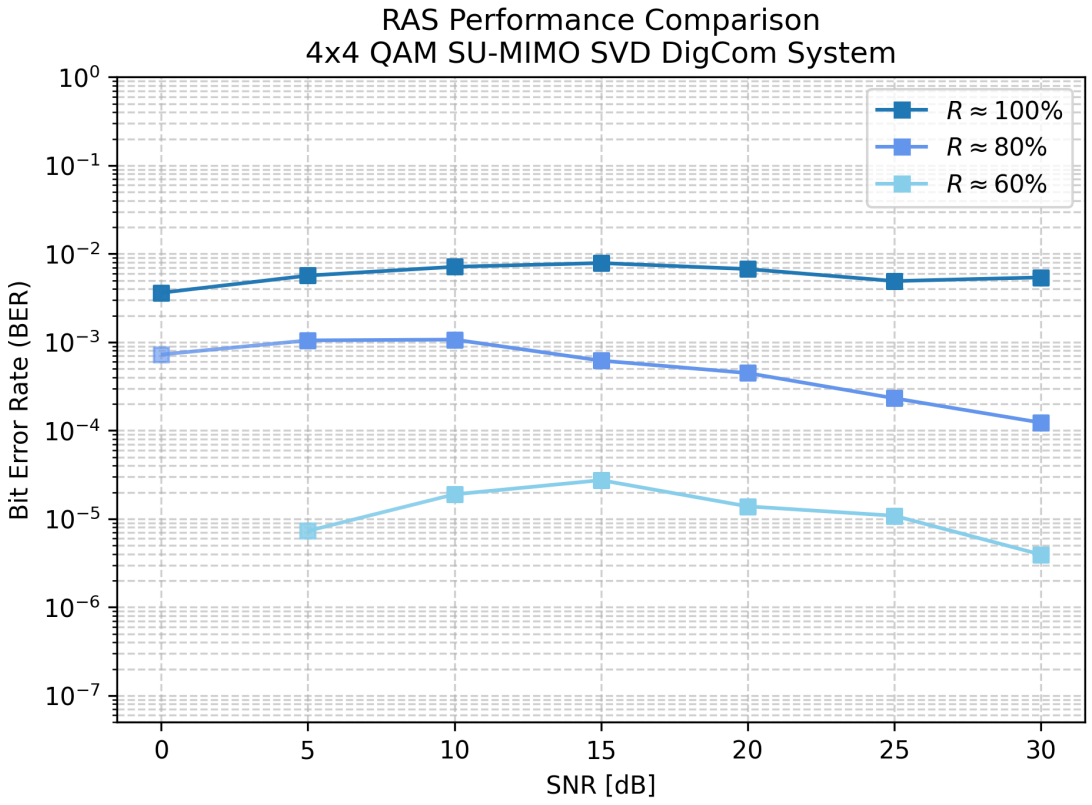
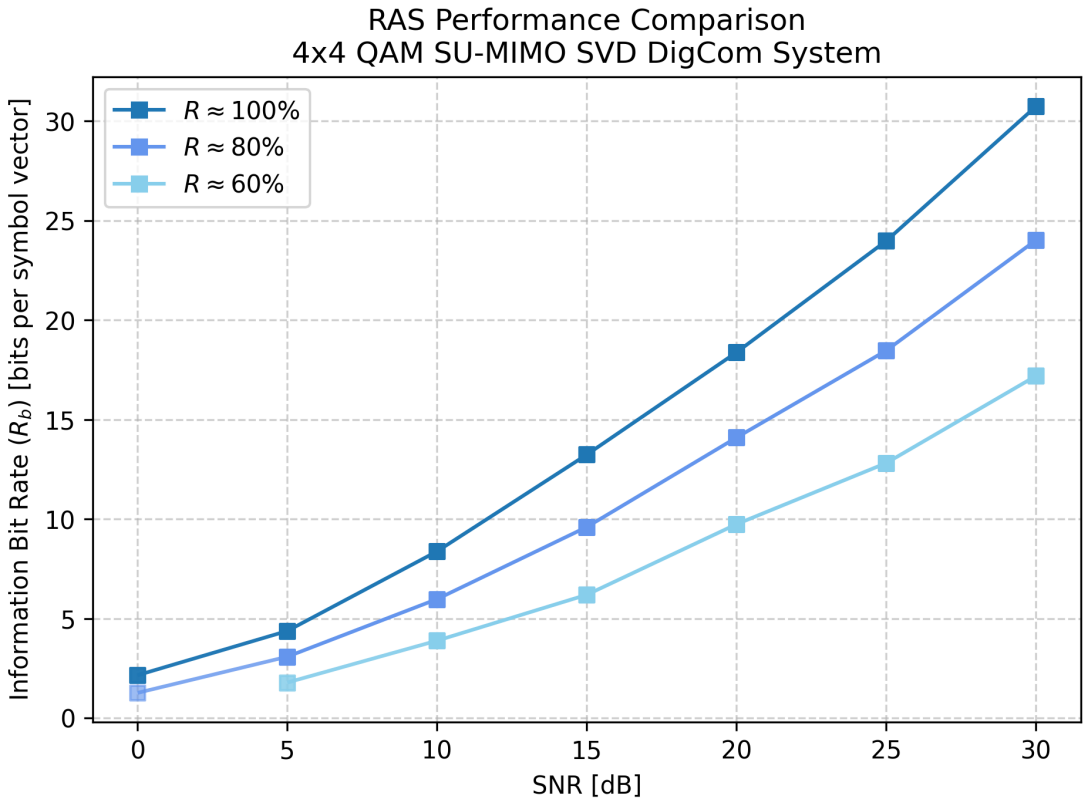
Simulation Results

Impact of **Resource Allocation Strategy** on BER & IBR Performance





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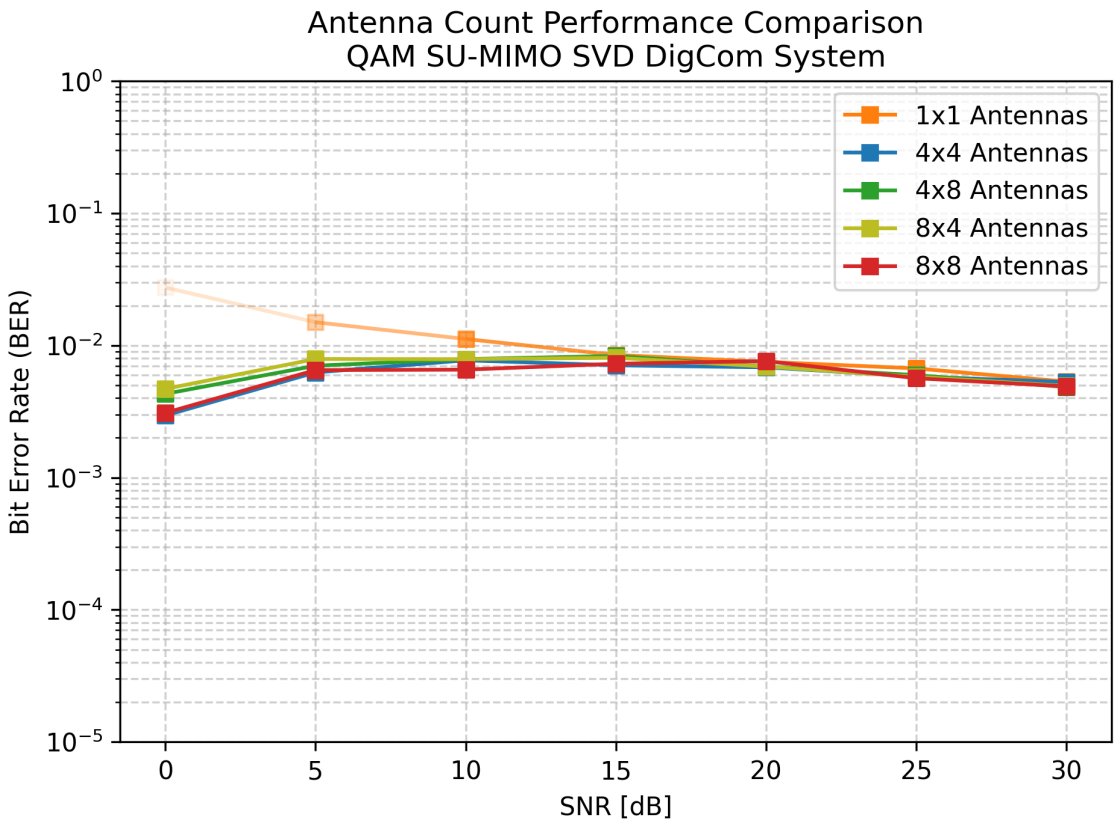
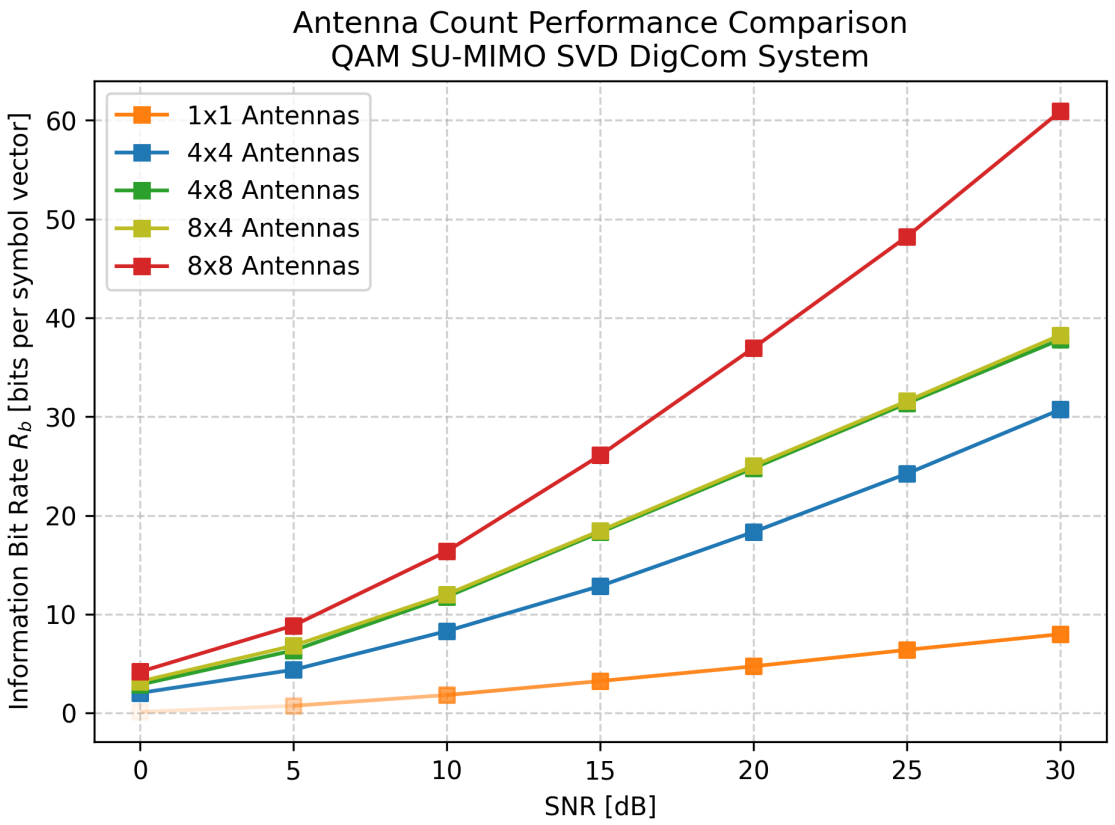
Simulation Results

Impact of **Antenna Count** on BER & IBR Performance





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Analytical Results

The bit error rate of the system (BER) can be calculated as the weighted average of the bit error rates of each eigenchannel of the system (BER_i):

$$\mathrm{BER} = \frac{1}{R_b} \sum_{i=0}^{\mathrm{R_H}} R_{b,i} \cdot \mathrm{BER}_i$$

The Bit Error Rate of each eigenchannel of the system (BER_i) equals the average amount of bit errors per received data symbol:

$$\mathrm{BER_i} \leq \frac{1}{R_{b,i} \cdot 2^{R_{b,i}}} \sum_{(\hat{\alpha},\alpha)\in\mathcal{C}_i^2} \mathrm{N}(\hat{\alpha},\alpha) \cdot \mathrm{Q}\left(\sqrt{\frac{\mathrm{P_i}\,\sigma_\mathrm{i}^2}{2\mathrm{N_0}}} \cdot |\hat{\alpha} - \alpha|\right)$$

Union Upper Bound:

$$\mathrm{BER_i} = \frac{1}{R_{b,i}} \cdot \sum_{(\hat{\alpha},\alpha)\in\mathcal{C}_i^2} \mathrm{N}(\hat{\alpha},\alpha) \cdot \mathrm{P}[\hat{a}_i = \hat{\alpha} \wedge a_i = \alpha]$$

Nearest-Neighbors Approximation:

$$\mathrm{BER_i} \approx \frac{\mathrm{K}}{R_{b,i}} \cdot \mathrm{Q}\left(\sqrt{\frac{\mathrm{P_i}\,\sigma_\mathrm{i}^2}{2\mathrm{N_0}}} \cdot \mathrm{d_{min}}\right)$$

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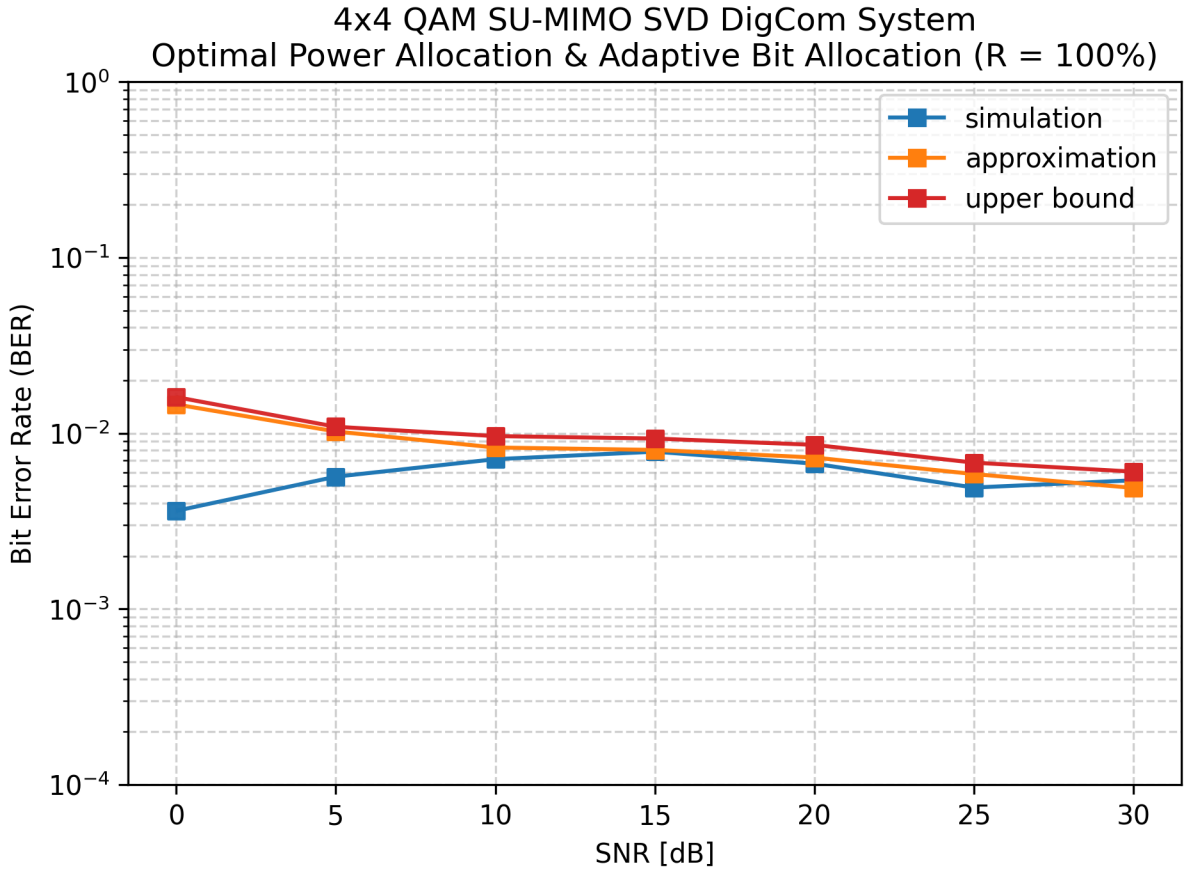
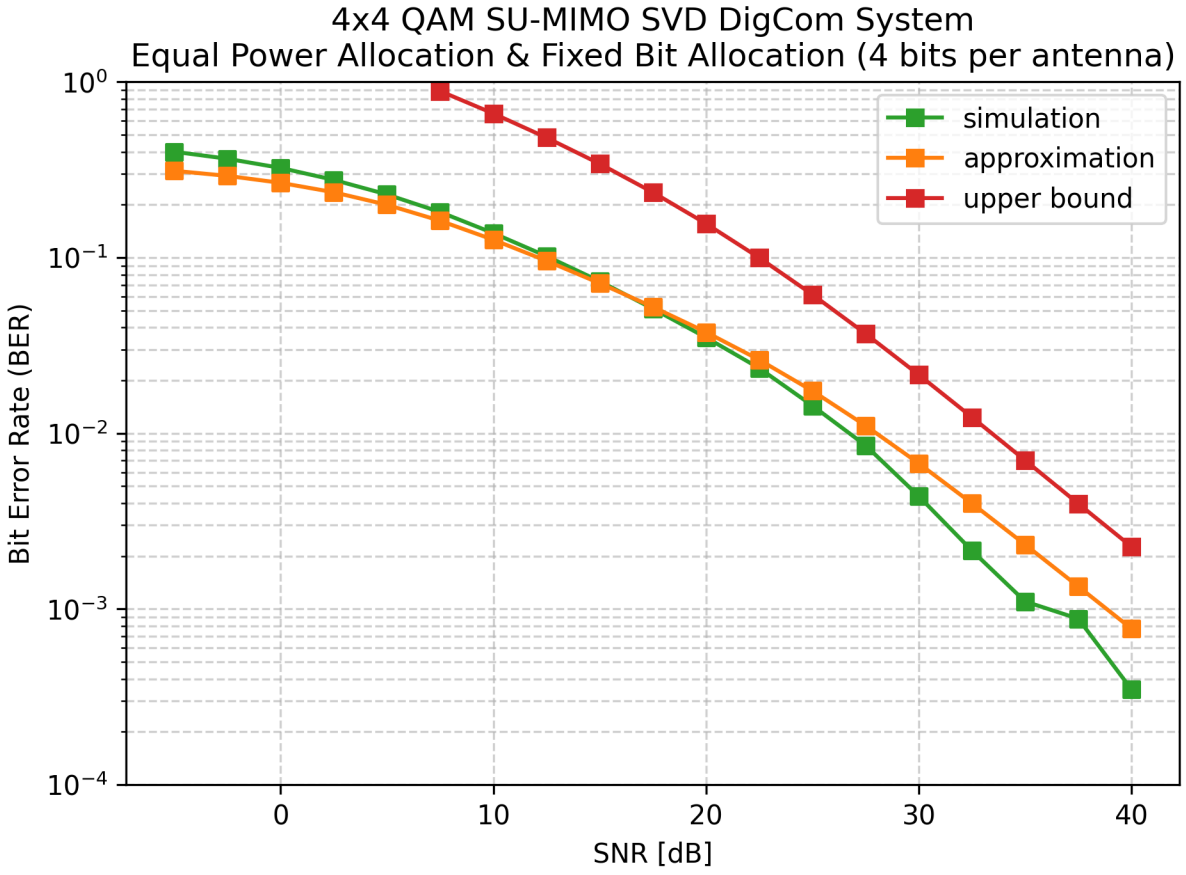
Analytical Results





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Conclusion

What are the specific **needs** and **restrictions** of your system in terms of **throughtput**, **reliability**, **complexity** and **hardware costs**? In which **environment** will your system be used?

- SNR ↑ → Throughput OR Reliability ↑
- Data Rate ↑ → Throughput ↑ BUT … Reliability ↓
- Antennas ↑ → Throughput OR Reliability ↑ BUT … Complexity AND Hardware Cost ↓
- Use of CSI → Throughput OR Reliability ↑ BUT … Complexity ↓
- …

→ Choose a system configuration that meets your needs and restrictions!

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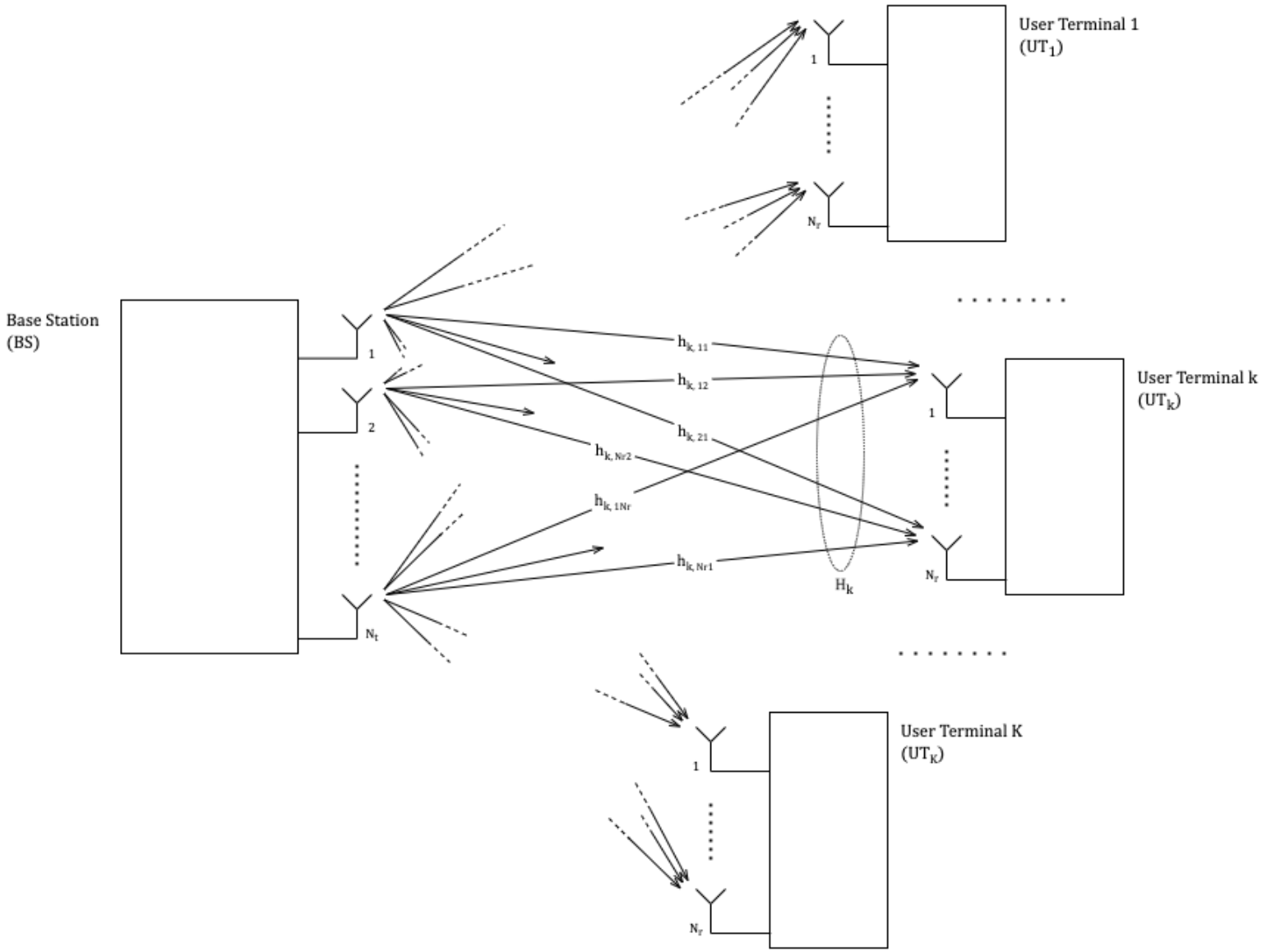
MU-MIMO
Precoding Techniques

Introduction



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System Model

- The transmitted signal $\mathbf{x}$ is constructed at the BS by allocating power to the data symbols $\mathbf{a}$, followed by precoding:

$$\mathbf{x} = \sum_{k'}^{K} \mathbf{F}_k \cdot \mathbf{P}_{k'} \mathbf{a}_{k'}$$

- The received signal $\mathbf{y}_k$ at UT_k can be written in terms of the transmitted signal $\mathbf{x}$ as:

$$\mathbf{y}_k = \mathbf{H}_k \cdot \mathbf{x} + \mathbf{n}_k$$

- The received signal $\mathbf{y}_k$ at UT_k is combined to obtain the scaled decision variable $\mathbf{z}_k$:

$$\mathbf{z}_k = \mathbf{W}_k \cdot \mathbf{y}_k$$

$$\mathbf{z} = \mathbf{WHF} \cdot \mathbf{Pa} + \mathbf{Wn}$$

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Problem Formulation

Weighted Sum-Rate Optimization

- Achievable Rate for UT_k:

$$R_{\mathrm{k}} = \sum_{\mathrm{s}=1}^{\mathrm{N_s}} R_{\mathrm{k,s}}$$

- Weighted Sum-Rate:

$$\mathrm{WSR} = \sum_{\mathrm{k'}=1}^{\mathrm{K}} \nu_{\mathrm{k}} R_{\mathrm{k}}$$

$$\max_{\mathbf{F},\mathbf{W},\mathbf{P}} \mathrm{WSR}$$

$$\text{s.t. } \|\mathbf{F}\,\mathbf{P}\|_{\mathrm{F}}^2 \leq P_t$$

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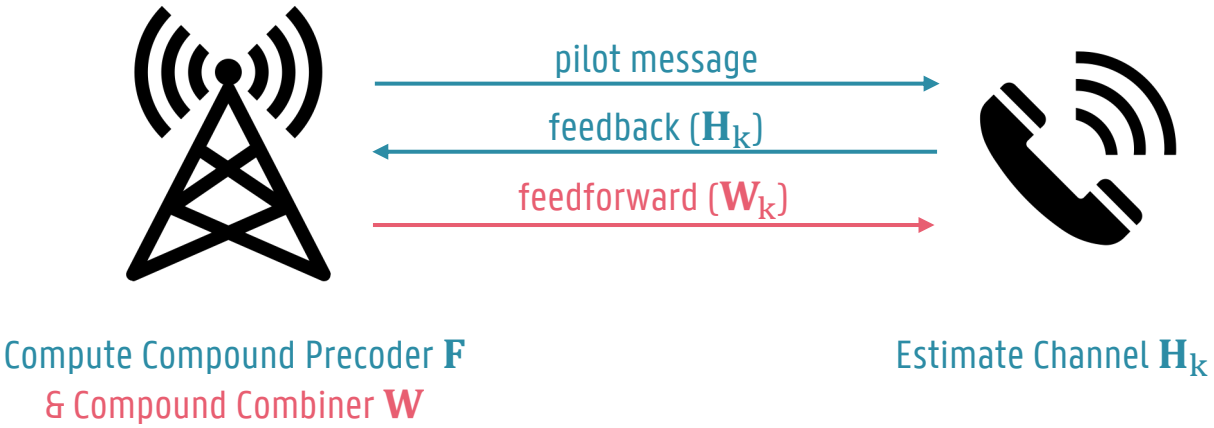
Precoding Techniques

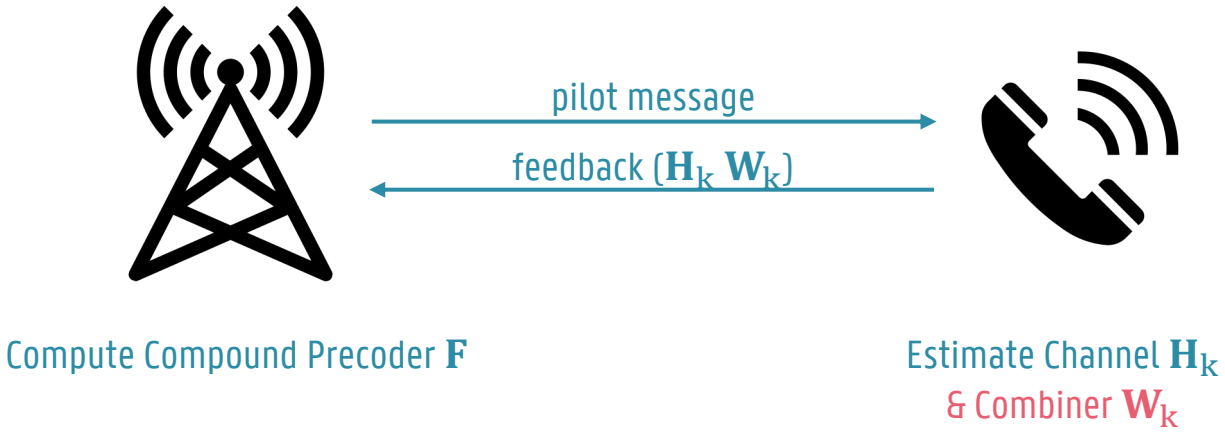
- WMMSE Beamforming (coordinated)
- WMMSE Beamforming (non-coordinated) with RSV Combiner
- ZF Precoder RSV combiner

Precoding Techniques

Coordinated VS Non-Coordinated



(a) Coordinated Beamforming



(b) Non-Coordinated Beamforming

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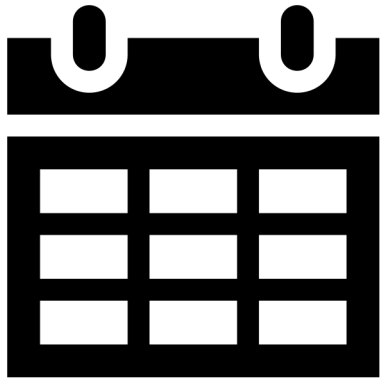
Future Plans

Future Plans

1. Simulation Analysis of MU-MIMO Precoding Techniques
2. Channel Model with Satellite Conditions
3. Impact of Channel Imperfections
4. Potential Benefits of Channel Compensation



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